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C03/AT1

1, Age and Photograph Preference

Ans

Given Data

Age	A	B	C
5-6	18	22	20
7-8	2	28	40
9-10	20	10	40

R-code

```
data <- data.frame (
  A=c (18,2,20),
  B=c (22,28,10),
  C=c (20,40,40)
)
cov (data$B, data$C)
cov (data)
cor (data$B, data$C)
cor (data)
```

Result

- * Covariance between B and C = -20
- * Correlation between B and C ≈ -0.189
- * The negative correlation is weak

To test the relationship between age and Photograph Preference:

```
tab <- matrix (c (18,22,20,
                  2,28,40,
                  20,10,40),
               nrow=3, byrow=TRUE)
```

chisq.test (tab)

Test gives approximately:

- * $\chi^2 = 29.60$
- * P-value $\approx 5.89 \times 10^{-6}$ * Since $P < 0.05$, there is significant relationship.

2)

Tennis player scores and Boxplot

Ans

Given data

12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23

R code

Points <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)

boxplot (points,

main = "Tennis Player scores",

ylab = "Points Scored")

Summary (points)

Result

Minimum = 10

Q1 = 13

Median = 16

Q3 = 20

Maximum = 35

$$IQR = 20 - 13 = 7$$

Upper limit:

$$20 + (1.5 \times 7) = 30.5$$

Since $35 > 30.5$, the score 35 is an outlier.

Conclusion

The boxplot shows that most players scored between 10 and 23 points. Player J, with 35 points, is an outlier and should be investigated.

Bank Loan Approval Using Decision Tree

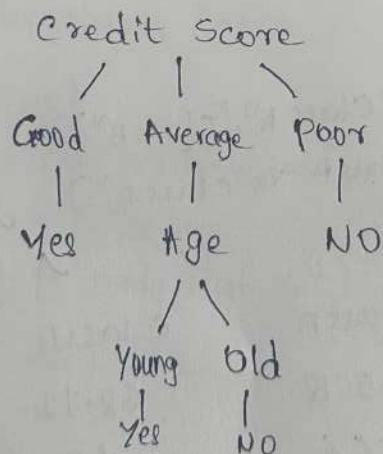
The dataset contains Age, Income, Employment status, Credit Score and Loan Approval.

R-code

```
library(rpart)
library(rpart.plot)
model <- rpart(
  loan ~ Age + Income + Employment + Credit,
  data = data,
  method = "class"
)
rpart.plot(model)
```

Decision Tree Concept

A simplified structure is:



Interpretation

- * Customers with good credit are more likely to receive a loan.
- * Customers with poor credit are likely to be rejected.

Conclusion

The decision tree provides a simple method for predicting loan approval. Credit Score is an important factor in dataset.

4) Comparison of class A and class B

Ans

Given

Class A

76, 35, 47, 64, 95, 66, 89, 36, 84

Class B

51, 56, 84, 60, 59, 70, 63, 66, 50

R-code

A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)

B <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)

mean(A)

median(A)

diff(range(A))

mean(B)

median(B)

diff(range(B))

boxplot(A, B,

names=c("Class A", "Class B"),
main="Class A vs Class B")

Result

Measure	Class A	Class B
Mean	65.78	62.11
Median	66	60
Range	60	34

Inference

- * Class A has a higher mean
- * Class A has a higher median
- * Class A has a larger range
- * Class B has more consistent scores.

Age and Body Fat

R code

age <- c(20)

fat <- c(80)

mean(age)

median(age)

sd(age)

mean(fat)

median(fat)

sd(fat)

boxplot(age,
main="Age Boxplot",
ylab="Age")

boxplot(fat,
main="Body Fat Boxplot",
ylab="Body Fat %")

plot(age, fat,
main="Age vs Body Fat",
xlab="Age",
ylab="Body Fat %")

qqnorm(age)

qqline(age)

qqnorm(fat)

qqline(fat)

Conclusion :-

The statistical measure describe the central tendency and variation of age and body fat.

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